

In order to achieve these objectives, the environmental education should be a continuous process and should be provided for all age groups at all levels, both in and out of school education.

Principles of Environmental Education

The principles of environmental education are as follows:

1. To appreciate the gifts of the nature.
2. To help acquire knowledge of the immediate environment.
3. To help understand the biotic and abiotic environment.
4. To help understand the inter-dependence of life at different trophic levels.
5. To help develop interest in flora and fauna of the environment.
6. To help develop interest in the community and the problem of the people and society.
7. To develop tolerance towards different casts, creeds, races, religions and cultures.
8. To help understand the effect of unchecked population growth or unplanned resource utilization of the world.
9. To love the neighbours and mankind as a whole.
10. To value equality, liberty, fraternity, truth and justice.
11. To examine trends in the growth of population and interpret them for the socio-economic development of the country.
12. To evaluate the utilization of physical and human resources and to suggest remedial measures.
13. To help diagnose the different causes of environmental pollution and to suggest remedial measures.
14. To help diagnose the causes of social tensions and to suggest methods for avoiding them.
15. To promote the value and necessity of local, national and international cooperation in the prevention and solution of environmental problems.

Stages of Environmental Education

Environmental education can be divided into two major categories : formal and non-formal.

1. Formal Environmental Education

It includes teaching and training of students of primary, secondary, higher secondary and university levels as well as teachers and environmental professionals.

(i) At **primary school stage**, students are sensitized about the environment through audio-visual and field visits.

(ii) At **lower secondary stage**, stress is on real life situations conservation and sustainable development. Teaching strategy includes those used at primary level along with general science, teaching, practicals and field work.

(iii) At **higher secondary school stage**, conservation knowledge, problem identification and action skills are highlighted. Emphasis is given on teaching, practicals and field work.

(iv) At **college stage**, maximum stress should be on knowledge regarding sustainable development. Students should be taught about conservation, real life situations and awareness contents should include science based teachings, practicals and field work.

(v) At **university stage**, stress is given on teaching research and extension. Some research institutions, engineering colleges and IIT's offer courses in *environmental studies*. At university stage, there are four major areas of environmental education *viz.* environmental engineering, conservation and management, environmental health and social ecology.

2. Non-Formal Environmental Education

It deals with youths and adults from all segments of the society such as family, factory workers, managers, decision makers in environmental as well as non-environmental fields.

The non-formal environmental education is being imparted in our country in the following ways:

- (i) By celebrating the world environment day (*i.e.* on 5th of June).
- (ii) By organising eco-development camps for youth to create awareness about basic ecological problems.
- (iii) By imparting in service and pre service training to the professionals, technical persons and legal experts.
- (iv) By arranging regular courses for senior executives and administrators.
- (v) Tribals are being educated about their environment by involving their leaders, women and youth.
- (vi) Adult education programmes also include education on the environment and its conservation.
- (vii) There are foundation courses for probationers, IAS, IPS, IFS and Armed forces.
- (viii) There are forums for public representatives *i.e.*, for M.P's and M.L.A.'s to discuss environmental problems.
- (ix) Many NGO's (Non-government organisations) are actively involved in the programmes like environmental education, conservation of nature, pollution control and afforestation.

Importance of Environmental Education

Environment belongs to all and is thus important for all, whatever be the occupation or age of a person, he or she will be affected by environment and will also affect the environment by his or her deeds. The main importance of environmental education are as follows:

1. Environmental education is important for the economy and welfare of human society.
2. It helps us in careful handling the issues like pollution menace, over exploitation of natural resources, food security and sustainable development.
3. In this age of science and technology, eco-friendly technology or eco-technology is the need of hour, because it promotes blending of the traditional knowledge and technology with modern technologies. It can be achieved only through proper environmental education.
4. Recless use of agrochemicals has degraded the environment and has disturbed the ecological balance. Environmental education helps us to find ways and means to maintain the ecological balance.
5. Environmental education demonstrates how man can derive benefits from the environment without destroying it.
6. Environmental education trains us to conserve our fast depleting natural resources.
7. Environment has been a source of happiness for man and time has come to preserve this happiness for man by imparting environmental education.

ecosystem brought about by natural process is counter balanced by changes in other components of the environment.

2. **Man-made or Anthropogenic Environment.** The environment which has been modified by human activities is called man-made environment. Man is the highest of all the creatures on this earth. He is modifying the environment according to his own needs and ways without taking into account its consequences. Increase in the scientific technologies which are the product of human brain is now deteriorating the environment.

HUMAN INTERFERENCE WITH THE ENVIRONMENT

Every living species of plants or animals influences its environment and in turn gets influenced by it. The magnitude of such influences is not usually high in these species because of the fact that due to natural checks their population cannot rise beyond certain limits and they can also not modify their own way of life. However, man is an exception. With increasing scientific knowledge, man is able to modify the environment to suit his immediate needs much more than any other organism. This enables man to improve the quality of his life.

Since the very beginning of human civilization, some thousand years ago, man started interfering with the environment. He devastated forests by cutting trees for the wood and for other household needs. He removed stretches of forests for bringing land under cultivation. He killed animals—the gentle ones, for food and the fierce ones, for safety. He had polluted the rivers with chemicals from factories, thereby making the water unfit for his needs. All these activities, however did not affect the environment too seriously upto a fairly long time, because the population was not too high and the life style was not so complex. The natural self-purifying and self-cleansing capacity of the environment was undeteriorated.

However, with the scientific and industrial revolution in the recent past, there has been immense impact of man on his environment. Man has failed to realise that any change upsets the balance of the ecosystem as a whole. All the devastating effects of man's effect to control nature have occurred because he has upset the balance relationship of the organisms that make up the environment. Huge industrial installations every year, introduction of faster mode of transport and sprouting up of large crowded cities (urbanization) are the main outcomes of the modern civilization. These and a large number of many others are contributing to what is called environmental pollution. An example of which is also the widespread use of insecticides. The immediate effect was a reduction in the population of pests and an increase in the yield of crops. But these insecticides also poisoned and killed birds which feed on insects. As a result, the next generation of pests multiplied even faster in the absence of their natural enemies, and damage to crops was much more. Increasing industrialization is also causing much danger to man's life by polluting the environment.

NEED FOR PUBLIC AWARENESS

With the advancement in technology, explosive increase in human population, scarcity of space and food, deterioration of hygienic conditions, depletion of natural resources and socio-economic problems, one needs extensive and exhaustive study of the environment, particularly in relation to human survival and benefit. The subject is gaining more popularity with its name as **environmental biology** or **environmental studies**. However, the environmental biology is not a new discipline but simply an extension of ecological approach which stresses upon the study of environment in its totality with special emphasis on the welfare of man.

8. Environmental education helps in inculcating attitudes and values, conducive to environment protection and understanding of inter-dependence of nature and man.
9. It creates an increased civic sense and an awareness of the economic, political and ecological inter-dependence of the modern world.
10. It helps to understand different food chains and the ecological balance in nature.
11. It directs attentions towards the problem of population explosion, depletion of natural resources and the pollution of environment and the methods of solving such problem.

Scope of Environmental Education

Environmental studies is very important and has following scopes:

1. There is a need for trained manpower at every level to deal with environmental issues like safe and clean drinking water, hygienic living conditions, clean and fresh air, fertile land, healthy food and sustainable development.
2. Environment protection and management, environmental laws, business administration and environmental engineering are emerging as new career opportunities.
3. Since the pollution control laws becoming more stringent, industries need environment experts to control pollution and disposal of wastes.
4. Environmental experts are now in great demand in industries for adopting green technologies to reduce pollution and to cut down the costs of effluent treatment.
5. These days, there is huge market world over for waste disposal and pollution control technologists and environmental experts.
6. If we want to live in a clean, healthy, aesthetically beautiful, safe and secure environment for a long time and wish to hand over a clean and safe earth to our children, we will have to include environmental experts in all developmental and policy making committees.

IMPORTANT TERMS

1. **Adaptation.** Structural and functional characteristics of organisms which enable them to survive and reproduce in a particular environment.
2. **Anthropogenic.** Human generated ; caused by humans.
3. **Atmosphere.** The gaseous mantle that envelops the earth surface.
4. **Climate.** Average weather of an area including general pattern of atmospheric seasonal variations and weather extremes.
5. **Ecology.** The study of reciprocal relationship between organisms and their environment.
6. **Edaphic.** Related to soil.
7. **Environment.** Surroundings in which organisms live.
8. **Habitat.** Place occupied by an organism/population/community which is exposed to particular combination of environmental factors.
9. **Microclimate.** Represents to climatic conditions in the areas of limited size.
10. **Sustainable Development.** Increase in standard of life that can be maintained over a long term without degrading the environment or compromising the ability of future generations to meet their own needs.
11. **Weather.** Refers to the short term properties of the atmosphere at a given place and time.